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Modelling and application of joint maintenance grouping and workload smoothing for an automotive plant

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ABSTRACT

In the maintenance optimisation framework, grouping maintenance is a promising solution for maintenance planning of multi-component systems, in which maintenance activities are performed together to reduce maintenance costs. One of the most widely identified challenges in real applications of grouping maintenance is that it may disturb the maintenance workload balance (smoothness), causing many difficulties in production and/or labour scheduling and inventory management. In this study, we propose a joint optimisation approach for maintenance grouping and workload balancing to address the above challenge. First, a mathematical model of the joint optimisation problem was derived. A multi-objective grouping optimisation approach based on the Weighted Sum model and Genetic Algorithm was implemented to determine the Pareto-optimal grouping solution. The proposed approach was applied to a real case study of an automotive plant comprising 40 production lines with 1090 components. The results highlighted the advantages, effectiveness, and flexibility of the proposed maintenance approach in real-world applications.

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Maintenance grouping; workload smoothing; automotive maintenance; multi-objective genetic algorithm; multi-component system

1. Introduction

The maintenance cost is an important part of the total operation cost and can constitute 15–40% of the production cost, depending on the manufacturing industry (Van Tung and Yang 2009). Therefore, maintenance optimisation has been intensively investigated to reduce maintenance costs and improve system performance (system reliability, energy efficiency, etc.) (A. Sharma, Yadava, and Deshmukh 2011; Dekker 1996; Ding and Kamaruddin 2015). In the past decades, several maintenance models have been proposed and can be classified into three types: individual maintenance (IM), opportunistic maintenance (OM), and grouping maintenance (GM) (Vu et al. 2020). In IM, maintenance activities are separately and independently planned for each component according to its characteristics. For example, a component can be preventively replaced after a specific operational time (age-based replacement model) (Nakagawa and Kowada 1983). OM considers the possibility that the maintenance of a component may be advanced to utilise maintenance opportunities, such as system downtimes and maintenance cost discounts (Ab-Samat and Kamaruddin 2014; Thomas, Levrat, and Iung 2008). Because maintenance opportunities usually occur randomly over time, the main drawback of the OM is that the

maintenance plan is almost unknown in advance, leading to many difficulties in logistical support (maintenance crew availability, spare parts ordering, transportation, and technician travel). The GM considers the possibility that the preventive maintenance (PM) of several components may be jointly executed simultaneously (Vu et al. 2020), that is, only the PM of one component is considered as an opportunity to execute the PM of the other components (Do et al. 2013; Wildeman, Dekker, and Smit 1997). GM logistics support is less challenging than OM logistics support because the grouping plan may be known in advance. GM is a promising solution for the maintenance planning of multi-component systems, and recently, it has been intensively developed (de Jonge and Scarf 2020; Nowakowski and Werbińska 2009).

Existing studies on GM provide promising solutions for maintenance modeling, management, and optimisation. However, several practical issues must be considered (Khatab et al. 2017; Koochaki et al. 2013). To the best of our knowledge, workload balance, and maintenance sequences have not been considered for GM optimisation and maintenance decision-making. A maintenance plan is usually optimised with respect to the requirements of the system/components. The global maintenance plan of the system may be unbalanced, that is, the

number of maintenance activities (workload) in specific periods may differ significantly from that of others. This situation is worse when GM is applied because maintenance activities are grouped. In addition, the sequence of maintenance activities may need to satisfy the specific requirements and references of the system or maintenance process. This unbalanced maintenance workload and disorganised maintenance sequence result in many difficulties in maintenance support, specifically labour (maintenance personnel) schedules and spare parts management. Workload balancing/smoothing and maintenance sequence constraints should be considered when grouping maintenance to improve the feasibility of grouped maintenance plans. This scientific issue motivated us to develop a joint optimisation approach for maintenance grouping and workload balancing. **In addition, this scientific issue is strongly linked to a real use case (an automotive maintenance workshop) provided by our industrial partner.** To address this challenging issue, we first developed a mathematical model for the joint optimisation problem. The maintenance sequence was considered a constraint for the proposed model. To solve the NP-complete optimisation problem and determine the Pareto-optimal grouping solution (which balances the maintenance cost and workload smoothness), a multi-objective optimisation based on the Weighted Sum model and Genetic Algorithm (WSGA) was adopted. The WSGA is an adequate solution to the problem considered because it represents the relative importance of the objectives (Jaigirdar et al. 2022; Marler and Arora 2010; R. Sharma, Kodamana, and Ramteke 2022). Compared with multi-objective optimisation meta-heuristics such as NSGA (Nondominated Sorting Genetic Algorithm), NPGA (Nondominated Ranked Genetic Algorithm) and MOEA (Multi-Objective Simulated Annealing) (Luiz Junho Pereira et al. 2022; Wang and Liu 2014; Zhou et al. 2011), WSGA transforms the functions to have similar magnitudes, and do not naturally dominate the aggregate objective function. This can help accurately set the weights to reflect preferences (I. Y. Kim and Weck 2006; Marler and Arora 2010). In addition, the WSGA allows for the quantification of the importance of one objective to another, which is helpful for maintenance operators and decision-makers. Bressi, Santos, and Losa (2021) and Chen et al. (2017).

To evaluate the performance of the proposed approach, a real case study of an automotive maintenance workshop—which supports the maintenance of an automotive plant composed of 40 production lines with 1090 components—was conducted. The results highlighted the advantages, effectiveness, and flexibility of the proposed approach. Furthermore, they emphasised the importance of considering the workload balancing and maintenance

sequence constraints when making GM decisions (Jaigirdar et al. 2022; Marler and Arora 2010; R. Sharma, Kodamana, and Ramteke 2022).

The remainder of the paper is organised as follows: Section 2 presents a short literature review on GM and workload balancing to underline the contributions of this study. Section 2 describes the system considered, its maintenance activities, and specific assumptions. A joint optimisation problem is introduced in Section 3. Section 4 presents the mathematical model of the problem. Section 5 presents an optimisation algorithm based on the WSGA to determine the best grouping solution. In Section 6, the performance of the proposed approach is validated using a case study involving an automotive company. Finally, this work's conclusions and future research directions are reported in Section 7.

2. Literature review on grouping maintenance and workload balancing

GM aims to group the execution of different PM activities to reduce the total maintenance costs. GM models can be categorised into stationary GM models (infinite horizons), dynamic GM models (finite horizon), and dynamic GM models based on the rolling horizon (Nicolai and Dekker 2008).

Stationary GM models provide static rules that do not change over a planning horizon. These fixed rules can be based on the maintenance frequency for groups of related maintenance activities or can be obtained using several control-limit strategies. For example, Goyal and Kusy (1985), and Dekker, Frenk, and Wildeman (1996) proposed a stationary GM model in which each component i ($i = 1, \dots, n$) of a series system has an optimal PM cycle in the form $k_i \cdot T$, where $T > 0$ is a real number, k_i is an integer, and n is the number of components in the system. **The decision variables T and k_i are determined by minimising the average maintenance cost rate of the system over an infinite horizon.** The number of variables to be optimised increases with the number of components in the system. Vu, Do, and Barros (2015) proposed a stationary GM model for redundancy structure systems. According to this model, the system is maintained at every τ units of time. At each maintenance epoch, the components are selected to be maintained together based on their mean remaining lifetime (*MRL*) and their important role in the system structure. The parameters τ and *MRL* are optimised for the system's long-term maintenance cost rate. The number of parameters to be optimised does not depend on the number of components; however, similar to the other stationary GM models, the proposed model suffers from a significant drawback: the maintenance plan can not be updated to take advantage of

maintenance opportunities, such as system stops owing to component failures or production plans.

Dynamic GM models were initially introduced to change the maintenance plans to take advantage of maintenance opportunities by considering a short-term horizon (Nakagawa and Mizutani 2009; Wu et al. 2010). In that way, Wildeman, Dekker, and Smit (1997) proposed a dynamic GM model based on the rolling horizon. The model comprises two main steps: 1-individual maintenance optimisation and 2-grouping maintenance optimisation. The individual maintenance optimisation step aims to determine the optimal PM cycle for each component with respect to the long-term maintenance objective. In the second step, a short-term interval is considered. All the PM activities of the components in the interval are identified based on their optimal PM cycles. Thereafter, these PM actions are optimally grouped together to reduce the maintenance costs in the considered interval. The advantage of this grouping model is that the grouping plan can be updated over time (short-term grouping). Simultaneously, the long-term maintenance performance is guaranteed (grouping based on the individual maintenance plan, which is optimised in the long term). This model has drawn significant attention from researchers, and many studies have been conducted to improve its performance by considering structure redundancy (Vu et al. 2014a), maintenance duration (Vu, Do, and Barros 2018), predictive information (Chang et al. 2019; Do et al. 2013; Liang and Parlikad 2020; Van Horenbeek and Pintelon 2013), major accident risks (Okoh 2015), human factors (Sheikhalishahi, Azadeh, and Pintelon 2017), stochastic dependence (Fan et al. 2021; H. Li, Deloux, and Dieulle 2016), imperfect maintenance (Pargar, Kauppila, and Kujala 2017), production planning (Alimian, Ghezavati, and Tavakkoli-Moghaddam 2020), Particle Swarm Optimization (Chalabi et al. 2016; Hanini et al. 2022). The performance of the dynamic GM model has been proven through many real applications in production systems (van Dijkhuizen 2000), heavy vehicles (Bouvard et al. 2011), pipeline assets (F. Li et al. 2011), offshore wind turbines (Hameed and Vatn 2012; Lu et al. 2017), and mining industry (Viveros et al. 2021).

From a practical perspective, having a good PM model is essential; however, ensuring that the model is feasible in terms of maintenance support, including spare parts and limited technicians, is also important. Chen et al. (2017) considered the problem of assigning technicians to aircraft-maintenance tasks. The authors formulated the problem as a bi-objective optimisation model that minimises the total cost while simultaneously achieving fairness in workload allocation among the different available technicians. The same problem was considered in

Tavakoli Kafiabad, Zanjani, and Noureldath (2022) and Turan, Kosanoglu, and Atmis (2021). Assigning technicians to predetermined maintenance tasks is challenging. This leads to additional costs because the maintenance plan is optimised only with respect to the maintenance costs. Therefore, Safaei, Banjevic, and Jardine (2011) and Koochaki et al. (2013) introduced constraints related to the availability of external and internal technicians for the maintenance optimisation problem. Ertogral and Öztürk (2019) and Akl et al. (2022) studied the joint optimisation of PM frequency and the assignment of technicians. Joint optimisation guarantees that maintenance is optimally planned to reduce maintenance costs, and is suitable for maintenance support. However, these studies usually consider simple maintenance models, such as the periodic PM model, and the optimisation is only performed for short-term horizons. As mentioned in the previous section, for the GM models, the problem related to maintenance support is more challenging than that for the other PM models because the workload is usually unbalanced owing to the grouping of PM activities. However, maintenance support for maintenance grouping has rarely been investigated. Do et al. (2015) introduced a constraint on the number of technicians in dynamic GM models. Nguyen et al. (2019) developed a joint optimisation of maintenance and repair routing. Nevertheless, workload balancing and maintenance sequence constraints have not been investigated using dynamic GM models.

In summary, our problem setting is unique in terms of integrating workload balancing and maintenance sequence constraints with a dynamic GM model based on a rolling horizon. Further details regarding the problem setting are provided in the next section.

List of Abbreviations and Notations

PM	Preventive maintenance
CM	Corrective maintenance
GM	Grouping maintenance
MTBF	Mean time between failures of component
N	Number of production line
N_r	Number of repairmen in the maintenance team
n_j	Number of components/ maintenance operations of line j
Tb, Te	Beginning and ending dates of the planning horizon
Tb_k, Te_k	Beginning and ending dates of the k th period
W_k	Maintenance workload of the k th period
$\bar{w}$	Average maintenance workload in the planning horizon

M	Total number of PM actions within the planning horizon
ij	Component i of line j
m_{ij}	Number of PM actions of component ij
PM_{ij}^l	l th PM action of component ij
C_{ij}^p, C_{ij}^c	Preventive and corrective maintenance costs respectively of component ij
C_{ij}^d	Unavailability cost rate
d_{ij}	Preventive maintenance duration of component ij
S_j^c, S_j^p	Corrective and preventive set-up cost for maintenance of a component of line j
$\Phi_{ij}(x)$	Expected CM cost due to the failure of component ij under replacement cycle x
$CR_{ij}(x)$	Long-term average maintenance cost rate of component ij under replacement cycle x
λ_{ij}, β_{ij}	Scale and shape parameters of Weibull distribution associated to component ij
$z_{ij}(t)$	Failure rate of component ij at time t

3. General descriptions and assumptions

3.1. System description

Consider a production system with N independent production lines where line j ($j = 1, \dots, N$) comprises a set of components. In this study, to simplify the analyses, only the critical components of each line were considered to simplify further analysis. A component is considered critical for the function of a production line if (i) its failure causes a shutdown of the entire production line, and (ii) and its failure frequency without any preventive maintenance intervention is significant. Let ij denote the i th component of line j . A reliability block diagram of the production system is shown in Figure 1, where n_j denotes the number of critical components in line j . The production lines operate independently; that is, when a line is stopped for any reason, the other lines continue to work normally. However, the stoppage of a line interrupts its production and reduces overall system productivity. Corrective and preventive maintenance are often considered

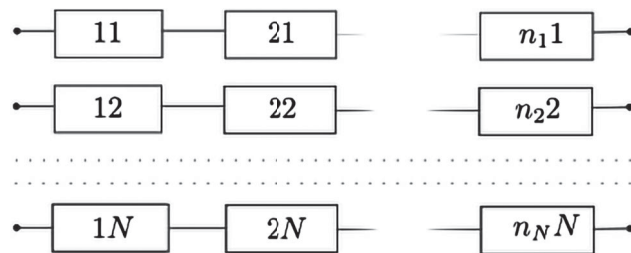


Figure 1. Reliability block diagram of a production system composed of N lines.

to maintain the performance of a production line and improve the productivity of the system.

3.2. Maintenance activities and costs

3.2.1. Corrective maintenance (CM)

As the failure of a component leads to a stoppage of its production line when the component fails; it is immediately repaired to restore the line to its operational state.

Corrective action restores the failed component to its state immediately before the failure (minimal repair). Failure is considered soft (minor) and can be fixed with less effort (M. J. Kim and Makis 2009; Rebaiaia and Aitkadi 2021). CM can be performed quickly at a low cost, and its duration can be neglected (an assumption made in this contribution). Leakage in water distribution systems, pipeline systems, loose cables, or screws can be considered examples of soft failures. Thus, when a corrective action is performed for component i of line j , the CM cost, denoted by C_{ij}^c , should be paid. C_{ij}^c is divided into two parts.

$$C_{ij}^c = S_j^c + c_{ij}^c \quad (1)$$

where,

- S_j^c is the corrective setup cost paid for different preparation actions (machine opening, maintenance site preparation, etc.) in line j . It is assumed that the setup cost does not depend on the component characteristics;
- c_{ij}^c is the corrective specific cost paid for maintenance resources, including workforce, maintenance tools, spare parts, and production loss owing to the failure of component i of line j .

3.2.2. Preventive maintenance (PM)

After a certain operational time, the failure frequency of the components increases and reaches an unacceptable level when only minimal repairs are applied. Therefore, preventive maintenance should be considered to avoid this situation. In this study, we assume that PM action is perfect, i.e. a PM action ensures the maintained component is in an 'as good as new' state (replacement action). Compared to a CM action, a PM action can occasionally be more complex and time-consuming. Therefore, the duration of PM must be addressed. Each PM action incurs a cost. Thus, the cost paid for the PM action of component i of line j can be expressed as follows:

$$C_{ij}^p = S_j^p + c_{ij}^p + D_{ij}^p \quad (2)$$

where,

- S_j^p is the preventive setup cost for preventive maintenance of component i in line j ;
- C_{ij}^p is the preventive-specific cost, including spare parts, workforce, and maintenance tool costs. As a PM action completely restores the maintained component, it is reasonable to assume that $C_{ij}^p > C_{ij}^c$;
- D_{ij}^p is the downtime cost owing to the production loss during PM maintenance of component i in line j . This downtime cost can be expressed as $D_{ij}^p = c_j^d \cdot d_{ij}$, where c_j^d and d_{ij} denote the downtime cost rate of line j and the maintenance duration of component i of line j respectively.

3.3. Specific assumptions

In addition to the above descriptions of the system and its maintenance activities, without deviating from our main objective, the following assumptions were made:

- (1) At instant t , a component is either in an operational or failed state. The component failure behaviour (time to failure) is described by a continuous probability distribution with an increasing failure rate;
- (2) The deterioration of components remains unchanged when the components are not functioning;
- (3) The components are stochastically independent; that is, the failure behaviour of a component does not depend on that of the other components;
- (4) Considering that the CM duration can be ignored, the maintenance workload is calculated primarily based on PM actions;
- (5) A maintenance team composed of N_r repairmen is responsible for the PM of all components in the system. The maintenance team is assumed to be able to perform all the required maintenance tasks during any maintenance period (weekly workload in our case study);
- (6) When several PM actions of production line j are jointly or consecutively executed, only one setup cost S_j^p must be paid because of the sharing of the setup cost between PM actions.

4. Problem statement and formulation

Based on the above assumptions, we describe the maintenance planning and workload smoothing problems in more detail. Furthermore, the need for the joint consideration of maintenance planning and workload smoothing in the maintenance optimisation framework is also highlighted.

4.1. Need of workload smoothing

Consider the planning horizon $PH = [Tb, Te]$, which is divided into K specific periods of time $PT_k = [Tb_k, Te_k]$ with $(k = 1, 2, \dots, K)$. For each period PT_k , the workload W_k is defined as the working time assigned to the maintenance team. Because the CM duration can be ignored, the workload is directly linked to the execution time of the PM actions during period PT_k and their duration. With a particular maintenance plan of the system in PT_k that defines the PM actions for specific components at specific times, the workload W_k can be calculated as:

$$W_k = \sum_{j=1}^N \sum_{i=1}^{n_j} \sum_{l=1}^{m_{ij}} IS(PP_{ij}^l, PT_k) \quad (3)$$

where,

- $PP_{ij}^l = [t_{ij}^l, t_{ij}^l + d_{ij}]$ is the l th PM period of component i of line j in a planning horizon PH and t_{ij}^l is the execution date of the l th PM action of component i of line j , denoted PM_{ij}^l ;
- m_{ij} is the number of PM actions of component i of line j in PH ;
- $IS(PP_{ij}^l, PT_k)$ is the duration of the intersection (overlap) between the two intervals PP_{ij}^l and PT_k .

$$0 \leq IS(PP_{ij}^l, PT_k) \leq \min(d_{ij}, d_{PT_k}), \quad (4)$$

with $d_{PT_k} = Te_k - Tb_k$ is the duration of PT_k .

Practically, PM actions are planned according to the components' needs (performing appropriate maintenance actions at the right time to avoid component failure at the lowest maintenance cost), which are not the same for different intervals PT_k and components. For example, component i of line j may not be preventively maintained in PT_k , but it may be maintained one or several times in PT_{k+1} because its failure frequency or deterioration level reaches an unacceptable level. Consequently, workload W_k varies over time.

If the planning horizon $PH = [Tb, Te]$ can be divided into K equal intervals, the duration of each interval is

$$d_{PT_k} = \frac{d_{PH}}{K}, \quad k = 1, \dots, K \quad (5)$$

where $d_{PH} = Te - Tb$ denotes the duration of PH . The variance in the workload in PH , denoted by V_W , can be measured as

$$V_W = \frac{1}{K} \sum_{k=1}^K (W_k - \bar{w})^2 \quad (6)$$

where $\bar{w} = \frac{1}{K} \sum_{k=1}^K W_k$ denotes the average maintenance workload in PH .

A high value of V_W indicates that the maintenance workload varies significantly in PH . This variation in maintenance workload directly causes many difficulties for maintenance logistics support, especially for: (i) personnel management (ensures that the maintenance personnel are sufficient for maintenance); (ii) spare part inventory (ensures the availability of spare parts at the time of maintenance); and (iii) maintenance budget management (ensures that the money is sufficient to pay for all maintenance actions in the considered period). Maintenance logistics support is usually designed based on the average maintenance workload value $\bar{w}$ and operates under real workloads W_k . The following unexpected situations may occur:

- W_k is higher than $\bar{w}$; maintenance resources are insufficient to ensure that all planned actions in PT_k are accomplished; Extra costs for renting external maintenance staff and money for emergency spare parts orders owing to out-of-stock costs must be paid;
- W_k is lower than $\bar{w}$ and maintenance resources are superfluous. This leads to workforce wastage, an increase in spare parts-holding costs, and an overestimated maintenance budget.

These two situations can be avoided by adjusting the maintenance plan such that V_W is minimal, that is, W_k is close to $\bar{w}$. The ideal case is obtained when the variance is zero and $V_W = 0$. The maintenance workloads at all intervals PT_k are equal to their average values ($\bar{w}$). Therefore, workload smoothing in maintenance planning is crucial for maintenance management and optimisation.

4.2. Joint optimisation problem

According to the above discussion, it is essential to adjust PM actions such that workload variance is minimal. However, if the PM is planned only for workload variance, it may not correctly satisfy the needs of the components. The maintenance plan may not be adequate, and the maintenance cost may be high because of (a) the loss of the components' useful life if the PM actions are planned earlier, (b) an increase in the production line/component failure frequency if PM actions are postponed. This challenging issue motivated us to develop a joint optimisation model for maintenance planning and workload smoothing (workload variance reduction). Our joint optimisation model is expressed as follows:

$$\underset{t_{ij}^l}{\text{minimize}} \quad \begin{cases} V_W(t_{ij}^l, K) \\ C(t_{ij}^l, PH) \end{cases} \quad (7a)$$

Subject to

$$Tb \leq t_{ij}^l \quad i = 1, \dots, n_j; j = 1, \dots, N; l = 1, \dots, m_{ij} \quad (7b)$$

$$t_{ij}^l + d_{ij} \leq Te, \quad i = 1, \dots, n_j; j = 1, \dots, N; \\ \times l = 1, \dots, m_{ij} \quad (7c)$$

$$IS(PP_{ij}^l, PP_{i'j'}^{l'}) = \emptyset, \quad l' \neq l, i' \neq i, j' \neq j \quad (7d)$$

$$t_{ij}^l + d_{ij} < t_{ij}^{l+1}, \quad l = 1, \dots, m_{ij} - 1 \quad (7e)$$

Thereafter, we determine the optimal execution dates t_{ij}^l that minimise both the workload variance V_W and the total maintenance cost C over the planning horizon PH (Equation (7a)). Constraints (7b) and (7c) ensure that the execution of the PM actions is strictly started and completed in PH . Considering that only one maintenance team is assumed, only one PM action is executed at a time; that is, the intersection between any two PM periods is null (Constraint (7d)). The last constraint, (7e), indicates that the $(l + 1)$ th PM action of a component can start only when its l th PM action is completed.

Although the workload variance (V_W) can be calculated directly using Equations (3) and (6), the calculation of the total maintenance cost (C) depends on the applied maintenance policy developed in the next section.

5. Grouping maintenance policy and cost evaluation

Before the workload variance and total maintenance cost in PH can be calculated, the maintenance policy used for maintenance planning must be defined. As mentioned in the Introduction section, three types of maintenance policies exist: individual maintenance (IM), opportunistic maintenance (OM), and grouping maintenance (GM). Considering that the maintenance plan provided by the OM is unknown in advance, it is difficult to determine workload variance. In addition, when compared to the IM, the GM can reduce the total PM setup and maintenance costs by maintaining several components simultaneously. However, joint maintenance also disturbs the maintenance workload. Therefore, it is important to consider GM for the joint optimisation problem presented in Subsection 4.2.

The GM policy proposed in Wildeman, Dekker, and Smit (1997) contains two main phases: **IM optimisation and GM optimisation**. The individual maintenance optimisation phase aims to determine all the individual PM actions of the components in PH with respect to the long-term maintenance cost rate objective. These identified PM actions are optimally regrouped to reduce maintenance costs during the GM optimisation phase. The advantage of this policy was that the grouping plan could

be updated over time (short-term grouping). Simultaneously, the long-term maintenance performance is guaranteed (grouping based on the individual maintenance plan, which is optimised in the long term). Next, we describe how the policy was developed to address this problem.

5.1. Individual maintenance optimisation

IM means that the maintenance of each component in the system is considered and optimised separately. We assume that component i of line j is preventively maintained every x_{ij} time units, and is minimally repaired when it fails. This policy is usually named 'periodic replacement with minimal repair at failures' in the literature (Do et al. 2013; Vu et al. 2020; Wildeman, Dekker, and Smit 1997). The replacement cycle x_{ij} is a decision parameter that is typically determined by minimising the long-term maintenance cost rate function. To develop the cost function, the Weibull distribution was used as an example to describe the component failure behaviour. Thus, the failure rate of component i of line j at time t , denoted $z_{ij}(t)$, can be described by the following equation:

$$z_{ij}(t) = \frac{\lambda_{ij}}{\beta_{ij}} \left(\frac{t}{\beta_{ij}} \right)^{\lambda_{ij}-1} \quad (8)$$

where λ_{ij} and β_{ij} are the scale and sharpness parameters of the Weibull distribution, respectively (Dekker 1996; Vu et al. 2014b). If proper lifetime data are available, that is, failure times are independent and identically distributed (i.i.d.), the parameters λ_{ij} and β_{ij} can be obtained using standard estimation methods, such as maximum likelihood estimation (Cohen 1965; Thoman, Bain, and Antle 1969). However, the i. i. d. assumption is not often validated in practical applications. Therefore, instead of directly using the parameters λ_{ij} and β_{ij} to calculate the failure rate $z_{ij}(t)$, we calculate the effective failure rate, denoted $z_{Eij}(t)$, based on $MTBF_{ij}$ (Mean Time Between Failures) and the aging parameter λ_{ij} as follows (Okoh (2015) and Vatn, Hokstad, and Bodsberg (1996):

$$z_{Eij}(x_{ij}) \approx \left[\frac{\Gamma(1 + 1/\lambda_{ij})}{MTBF_{ij}} \right]^{\lambda_{ij}} t^{\lambda_{ij}-1} \quad (9)$$

where $\Gamma(z) = \int_0^{+\infty} x^{z-1} e^{-x} dx$ denotes the gamma function. The main advantage of using $MTBF_{ij}$ and λ_{ij} is that the $MTBF_{ij}$ can be estimated from a non-homogeneous data set as the average of all recorded failure times, and λ_{ij} can be naturally obtained using the expert judgment approach (Noortwijk et al. 1992).

Based on the above effective failure rate, the expected cost that must be paid for the CM of component i of the

line j within interval $(0, x_{ij})$ is:

$$\Phi_{ij}(x_{ij}) = C_{ij}^c \cdot \int_0^{x_{ij}} z_{Eij}(t) dt = C_{ij}^c \cdot \left[\frac{\Gamma(1 + 1/\lambda_{ij})}{MTBF_{ij}} \right]^{\lambda_{ij}} \cdot \frac{1}{\lambda_{ij}} \cdot x_{ij}^{\lambda_{ij}} \quad (10)$$

By applying the renewal theory, the long-term maintenance cost rate denoted CR_{ij} , can be calculated as

$$CR_{ij}(x_{ij}) = \frac{C_{ij}^p + \Phi_{ij}(x_{ij})}{x_{ij} + d_{ij}} = \frac{C_{ij}^p + C_{ij}^c \cdot \left[\frac{\Gamma(1 + 1/\lambda_{ij})}{MTBF_{ij}} \right]^{\lambda_{ij}} \cdot \frac{1}{\lambda_{ij}} \cdot x_{ij}^{\lambda_{ij}}}{x_{ij} + d_{ij}} \quad (11)$$

The optimal replacement cycle, denoted x_{ij}^* , can be obtained by solving the following equation:

$$\frac{dCR_{ij}(x_{ij})}{dx_{ij}} \Big|_{x_{ij}=x_{ij}^*} = 0 \quad (12)$$

Based on the optimal replacement cycle x_{ij}^* , the execution time of all PM actions of component i of line j in PH can be determined using the following recursive formula:

$$t_{ij}^l = \begin{cases} Tb + [x_{ij}^* - age_{ij}(Tb)] + D_j(l-1, l) & \text{if } l = 1 \\ t_{ij}^{l-1} + d_{ij} + x_{ij}^* + D_j(l-1, l) & \text{if } 1 < l \leq m_{ij} \end{cases} \quad (13)$$

where,

- $age_{ij}(Tb)$ is the age (cumulative operational time) of the component i of line j at time Tb ;
- $D_j(l-1, l)$ is the cumulative downtime of line j in interval $[t_{ij}^{l-1}, t_{ij}^l]$; $t_{ij}^0 = Tb$.

Figure 2 shows the execution time of some PM actions of the two components 1 and 2 of the first production line in PH. In this example, component 1 is maintained twice at t_{11}^1 and t_{11}^2 , and component 2 is maintained only once at t_{21}^1 . We have $m_{11} = 2$ and $m_{21} = 1$.

The total maintenance cost of the system in PH, when all PM actions are individually maintained at their nominal execution time t_{ij}^l , is

$$C_I(PH) = d_{PH} \cdot \sum_{j=1}^N \sum_{i=1}^{n_j} CR_{ij}(x_{ij}^*) \quad (14)$$

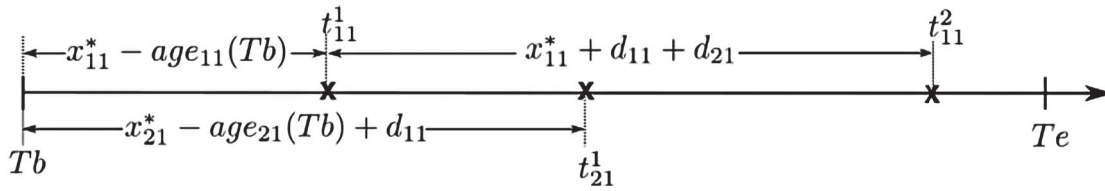


Figure 2. An example of the individual PM dates in PH.

5.2. Grouping maintenance optimisation

Without loss of generality, we assume that there are M preventive maintenance actions in the considered horizon PH . To reduce setup costs, the M preventive maintenance actions obtained in the previous phase should be grouped according to a grouping structure that is defined as a partition of the set of all M preventive maintenance actions.¹ Let us consider a grouping structure denoted GS , composed of U groups of PM actions $GS = \{G^1, G^2, \dots, G^U\}$. A group G^u ($u = 1, \dots, U$) contains n_{G^u} PM actions; that is, $\sum_{u=1}^U n_{G^u} = M$.

The maintenance team begins maintaining the group at t_{G^u} and finishes at $t_{G^u} + d_{G^u}$, where d_{G^u} denotes the total maintenance duration of G^u . To reduce the downtime of each production line during the grouping, it is preferred that the PM actions/tasks of the same line are executed simultaneously. The assignment of PM tasks to the available repairmen is simple when the number of PM tasks is less than or equal to the number of repairmen (N_r). When the number of PM tasks exceeds N_r , the assignment is complex and must be optimised to minimise the completion time. This problem is recognised as a makespan minimisation problem, a well-known problem in the optimisation domain, and is typically solved using mixed-integer linear programming (MILP). In this study, we adopted the MILP solution presented in Hashemian, Diallo, and Vizvári (2014) to determine the execution dates of each PM action in the group t_{ij}^u , maintenance duration of each line d_{ij}^u , and total maintenance duration of the group d_{G^u} . The details of this solution are presented in Appendix 1

From an economic point of view, grouping can help reduce maintenance duration, thereby downtime costs. Let ΔD_{G^u} denote the downtime cost savings during grouping. It can be calculated as follows:

$$\Delta D_{G^u} = \sum_{j \in G^u} c_j^d \cdot \left[\sum_{t_{ij}^u \in G^u} d_{ij} - d_{G^u}^j \right] \quad (15)$$

where $t_{ij}^u \in G_j^u$ indicates all the PM actions of line j in group G^u .

The joint maintenance of PM actions in G^u also yields a PM setup-cost reduction ΔS_{G^u} . According to the sixth

assumption in Subsection 3.3, when several PM actions of production line j are executed, only one setup cost S_j^p must be paid because of the sharing of the setup cost between PM actions. Let L^u denote the list of all the production lines with at least one component maintained in group G^u . The setup-cost reduction can then be calculated as follows:

$$\Delta S_{G^u} = \sum_{t_{ij}^u \in G^u} S_j^p - \sum_{j \in L^u} S_j^p \quad (16)$$

However, to be jointly executed at t_{G^u} , the optimal execution dates of the PM actions of component i in line j of G^u must be modified. This modification leads to a change in the expected repair costs in the two intervals $I_A = [t_{ij}^{l-1}, t_{ij}^l]$ and $I_B = [t_{ij}^l, t_{ij}^{l+1}]$. In particular, before the grouping, the execution date of PM_{ij}^l is at t_{ij}^l , the total operational time of component i of the line j in the two intervals I_A and I_B are equal to x_{ij}^* ; the total expected repair costs are then $M_{ij}(x_{ij}^*)$. After the grouping, the execution date of PM activity PM_{ij}^l becomes t_{ij}^u ; the operational time of the component in two intervals I_A and I_B are also changed to $x_{ij}^* + (t_{ij}^u - t_{ij}^l)$ and $x_{ij}^* - (t_{ij}^u - t_{ij}^l) - (d_{G^u}^j - d_{ij})$ respectively.

Grouping maintenance also leads to additional expected repair costs, denoted by ΔR_{G^u} .

$$\Delta R_{G^u} = \sum_{t_{ij}^u \in G^u} [\Phi_{ij}(x_{ij}^* + \Delta t_{ij}^l) + \Phi_{ij}(x_{ij}^* - \Delta t_{ij}^l - \Delta d_{ij}) - 2 \cdot \Phi_{ij}(x_{ij}^*)] \quad (17)$$

where, $\Delta t_{ij}^l = t_{ij}^u - t_{ij}^l$ and $\Delta d_{ij} = d_{G^u}^j - d_{ij}$.

The joint maintenance of the PM actions of G^u helps to reduce the maintenance cost compared with the case in which all PM actions are individually maintained at their optimal dates t_{ij}^l . This reduction is called the economic profit of G^u and denoted $EP(G^u, t_{G^u})$.

$$EP(G^u, t_{G^u}) = \Delta D_{G^u} + \Delta S_{G^u} - \Delta R_{G^u} \quad (18)$$

The optimal execution time $t_{G^u}^*$ for G^u can be determined by maximising the economic profit of $t_{G^u}^* = \arg \max_{t_{G^u}} EP(G^u, t_{G^u})$ (Wildeman, Dekker, and Smit 1997).

If $EP^*(G^u) = EP(G^u, t_{G^u}^*)$ denotes the maximum economic profit of G^u , the total maintenance cost of the system in PH when the PM actions are grouped according to the grouping structure GS can be indirectly calculated based on $C_I(PH)$ as follows:

$$C_G(GS, PH) = d_{PH} \cdot \sum_{j=1}^N \sum_{i=1}^{n_j} CR_{ij}(x_{ij}^*) - \sum_{u=1}^U EP(G^u) \quad (19)$$

The main goal of GM optimisation is to find the best grouping structure GS^* that minimises $C_G(GS, PH)$ in the horizon PH .

5.3. Adapted joint optimisation model for the grouping maintenance

When grouping maintenance is applied, the grouping structure must be optimised such that the total maintenance cost $C_G(GS, PH)$ and workload variance in PH are minimal. Therefore, we must express the workload variance in Equation (3) as a function of the grouping structure GS as follows:

$$W_k(GS) = \sum_{u=1}^U IS(PG^u, PT_k) \quad \text{with } 0 \leq IS(PG^u, PT_k) \leq \min(d_{G^u}, d_{PT_k}) \quad (20)$$

where $PG^u = (t_{G^u}^*, t_{G^u}^* + d_{G^u})$. The function used to calculate the workload variance in Equation (6) remains unchanged. The joint optimisation model can be rewritten as follows:

$$\begin{aligned} & \text{minimize}_{GS=\{G^1, G^2, \dots, G^U\}} \begin{cases} V_W(GS, K) \\ C_G(GS, PH) \end{cases} \quad (21a) \end{aligned}$$

Subject to

$$\max(m_{ij}) \leq U \leq M \quad \text{with } i = 1, \dots, n_j;$$

$$j = 1, \dots, N \quad (21b)$$

$$G^1 \cup G^2 \cup \dots \cup G^U = \cup\{PM_{ij}^l\} \quad \text{with } i = 1, \dots, n_j;$$

$$j = 1, \dots, N; l = 1, \dots, m_{ij} \quad (21c)$$

$$G^1 \cap G^2 \cap \dots \cap G^U = \emptyset \quad (21d)$$

$$0 < EP^*(G^u, t_{G^u}^*) \quad \text{with } u = 1, \dots, U \quad (21e)$$

$$\forall PM_{ij}^m \in G^u, \quad \text{if } \exists n \neq m \text{ then } PM_{ij}^n \notin G^u \quad (21f)$$

The decision variable of the problem is the grouping structure GS , which contains U groups G^u of PM actions. It must be optimised such that the maintenance cost and workload variance in PH are minimised. Additionally, the grouping structure must satisfy several constraints. Specifically, constraint (21b) indicates that the

number of groups in a grouping structure cannot be larger than the number of PM actions in PH . When $U = M$ indicates that all PM actions are executed separately (individual maintenance), U cannot be smaller than the number of PM actions of the same component in the horizon because multiple PM actions of a component in a group are prohibited. m_{ij} denotes the number of PM actions of component i of line j in PH , and we have $\sum_{j=1}^N \sum_{i=1}^{n_j} m_{ij} = M$. Constraints (21c) and (21d) ensure that GS is a disjoint union of the groups. The maintenance time of each PM action t_{ij}^l in G^u can be modified to be maintained jointly with the other actions in the group. However, it cannot exceed its limited interval PG_{ij}^l which is defined as:

$$PG_{ij}^l = [\max(Tb, t_{ij}^{l-1} + d_{ij}); \min(Te, t_{ij}^{l+1})] \quad (22)$$

Constraint (21e) ensures that all groups in GS are cost-effective. Finally, constraint (21f) ensures that a group is not composed of different PM actions for the same component.

To solve the above joint optimisation problem, multi-objective optimisation tools, such as the Weighted Sum method (I. Y. Kim and Weck 2006) and genetic algorithms (Vu et al. 2014b), are used.

6. Multi-objective grouping optimisation

In this section, a multi-objective grouping optimisation based on the Weighted Sum method and Genetic algorithm (WSGA) is proposed (see Figure 3) to solve the problem described by Equations (21a)–(21f). The WSGA contains two main steps: the Weighted Sum model (WS) and the Generic Algorithm (GA). The multi-objective maintenance grouping optimisation is first transformed into a mono-objective optimisation using the WS. The GA is applied to determine the ideal grouping structure that optimises the aggregated objective function with respect to different constraints from (21b) to (21f).

6.1. Weighted sum model

The weighted sum method is the most widely used method for multi-objective optimisation. This method transforms multiple objectives into an aggregated objective function by multiplying each objective function by a weighting factor and summing all weighted objective functions (I. Y. Kim and Weck 2006). Applying this method to our problem, we obtain the following aggregated objective function:

$$VC(GS, K, PH) = \alpha_v \cdot V_W(GS, K) + \alpha_c \cdot C_G(GS, PH) \quad (23)$$

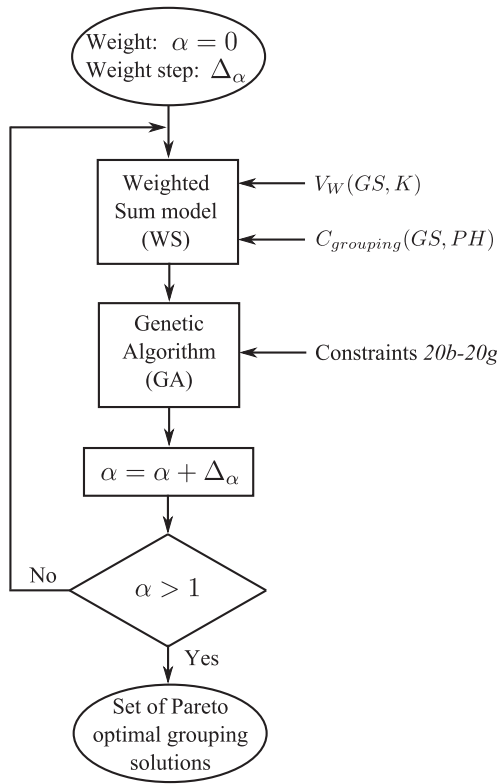


Figure 3. Multi-objective grouping optimisation based on Weighted Sum model and Genetic Algorithm (WSGA).

where α_v and α_c are the weighting factors for the workload variance and maintenance cost, respectively.

Thus, in this study, unity-based normalisation was applied to the two objectives (workload variance and maintenance cost) to bring their values within the range $[0, 1]$ (see Appendix 2 for more details). Let $\tilde{a}$ denote the normalised value of a , Equation (23) can be rewritten as:

$$\begin{aligned} \tilde{V}C(GS, K, PH) \\ = \alpha \cdot \tilde{V}W(GS, K) + (1 - \alpha) \cdot \tilde{C}_G(GS, PH) \end{aligned} \quad (24)$$

where $0 \leq \alpha \leq 1$ is the weighting factor for workload variance. Decision makers must select one value for weight α . However, selecting a weight is difficult and unfamiliar to many decision-makers. In practice, decision-makers usually select an acceptable solution from a set of Pareto optimal solutions that are non-dominated with respect to each other. Specifically, a solution GS is said to be a Pareto optimal solution if there is no other feasible solution GS' such that $\tilde{V}W(GS', K) < \tilde{V}W(GS, K)$ and $\tilde{C}_G(GS', PH) < \tilde{C}_G(GS, PH)$ simultaneously. Thus, identifying the set of Pareto optimal solutions is a key point for the decision-makers' selection of a compromise solution that satisfies all objectives. In this study, the set was determined by varying the value of α from 0 to 1 with a step size $\Delta\alpha$. If there are more than two

objectives, other multi-objective optimisation tools, such as a Non-dominated Sorting Genetic Algorithm (NSGA), should be more suitable (Konak, Coit, and Smith 2006).

6.2. Generic algorithm

Considering a specific value of α , the multi-objective grouping maintenance problem is converted into a mono-objective one. To obtain the optimal grouping structure denoted as GS^* , which minimises $\tilde{V}C(GS, K, PH)$ with respect to constraints (21b)–(21f), a genetic algorithm (GA) can be considered for the following reasons:

- The number of possible grouping structures increased rapidly with an increase in the PM actions. Determining the optimal grouping structure is an NP-complete problem (Vu et al. 2014b). The use of exact methods that consider all possible grouping structures requires excessive computation time. To tackle this NP-complete problem, an efficient global optimisation algorithm such as the GA is a promising solution that has been used in many studies (Vu, Do, and Barros 2018; Vu et al. 2014a, 2014b);
- The flexibility of the GA allows (a) to easily consider constraints (21b)–(21f); (b) can be applied to obtain the minimum or maximum value of the different objectives ($\tilde{V}C, \tilde{V}W, \tilde{C}_G$) with only slight modifications.

More details on the implementation process of the GA for solving our multi-objective grouping optimisation with constraints are presented in Appendix 3. In the next section, the performance of the proposed approach is examined using a case study of an automotive maintenance workshop.

7. Case study: an automotive plant

7.1. Description of the case study

The data used in the case study were obtained from an automotive maintenance workshop in France. The workshop supported the maintenance of $N = 40$ production lines of automotive engines. Because automotive engine production is a complex engineering process, each production line contains several components (motors, gearboxes, belts, pulleys, spindles, and machine tool axes). In this study, only the critical components that significantly impact the operation of each production line were considered. The number of critical components in each line n_j ($j = 1, \dots, 40$) ranged from 20 to 30. Data related to the mean time between failures $MTBF_{ij}$ (weeks), aging

Table 1. An extract of components' data, their maintenance cycles, and their maintenance cost rate.

Component ij	$MTBF_{ij}$	λ_{ij}	c_{ij}^p	C_{ij}^c	d_{ij}	x_{ij}^*	CR_{ij}
i j							
1 1	51,42	1,8958	15	26	0,25	51,3129	0,7627
2 1	25,71	1,7292	14	28	0,5	26,3009	1,5686
3 1	51,42	1,8958	15	26	2,5	51,1801	0,7610
4 1	12,85	1,5985	13	30	1	13,4209	3,2145
5 1	4,00	2,5149	14	19	0,25	3,6366	7,6572
...		...	...	...			
27 40	4,00	2,5149	14	19	5	3,6560	7,7192

parameter λ_{ij} , PM specific cost c_{ij}^p , CM cost C_{ij}^c , and PM duration d_{ij} (hours) of a critical component ij were collected, some of which are reported in Table 1. Note that by considering company confidentiality, the real data related to maintenance costs are scaled and provided in arbitrary cost units (acu). The setup cost for a PM action and the downtime cost rate of the production line j ($j = 1, \dots, 40$) are $S_j^p = 3$ (acu) and $c_j^d = 5$ (acu), respectively.

The components of the production lines are individually maintained with respect to their optimal replacement cycles x_{ij}^* . The optimal values of the replacement cycle and the average maintenance cost rate of all the components can be determined using the individual maintenance optimisation approach presented in Subsection 5.1. Table 1 lists examples. The maintenance of all the components of the system is ensured by a maintenance team containing 15 repairmen ($N_r = 15$).

Let us consider a one-year planning horizon PH with $d_{PH} = 52$ (weeks). If the components are individually maintained according to their optimal cycles in the planning horizon, the estimated maintenance cost in PH is $C_I(PH) = 662,880$ (acu), according to Equation (14).

The maintenance workload for each week was calculated based on individual maintenance dates. Based on Equation (6), the workload variance in PH was estimated to be $V_W^{individual}(PH) = 1,627,000$.

From these results, it is clear that individual maintenance leads to high maintenance costs and large workload variance. These limitations arise from the fact that when the maintenance of components is planned separately, the advantages of economic dependence between the components are not considered. In addition, the effect of an assigned PM action on the overall workload variance cannot be considered. The proposed grouping optimisation model was applied to reduce the maintenance cost and workload variance. The performance of the proposed model under different scenarios is discussed in the following subsections:

- The GM with respect to the workload variance objective ($\alpha = 1$) is denoted by the Workload GM model;

- GM with respect to both cost and workload variance objectives is denoted by the Bi-objective GM model ($\alpha = 0, 5$);
- GM with consideration of the maintenance order.

The performance of the proposed model was then compared with those of two other maintenance models:

- Individual maintenance (IM): Each component is replaced periodically based on the optimal PM cycle;
- Traditional Grouping Maintenance (GM): Grouping maintenance without considering the workload variance objective ($\alpha = 0$).

In addition, a performance comparison of the WSGA with the exact solution is presented in Appendix 5.

7.2. Mono-objective grouping maintenance planning

In this subsection, the maintenance cost and workload variance in the planning horizon PH are successively considered as objective functions for maintenance grouping optimisation.

7.2.1. Grouping maintenance planning with maintenance cost objective

Individual maintenance dates were grouped such that the maintenance cost was reduced significantly. Therefore, the weighting factor of the workload variance was set to $\alpha = 0$, and a GA was applied to solve the mono-objective grouping optimisation problem.

Figure 4 shows the grouping plan optimised with respect to the maintenance cost objective. In the grouping plan shown in Figure 4, the components are jointly maintained to reduce the setup cost. For example, most of the components are preventively maintained together at the 4th weeks. When we examined the overall grouping plan, no PM actions were executed separately. Compared to IM, GM yields a cost reduction, which is equal to

$$\frac{C_I(PH) - C_G^c(GS, PH)}{C_I(PH)} \cdot 100\% = 34\% \quad (25)$$

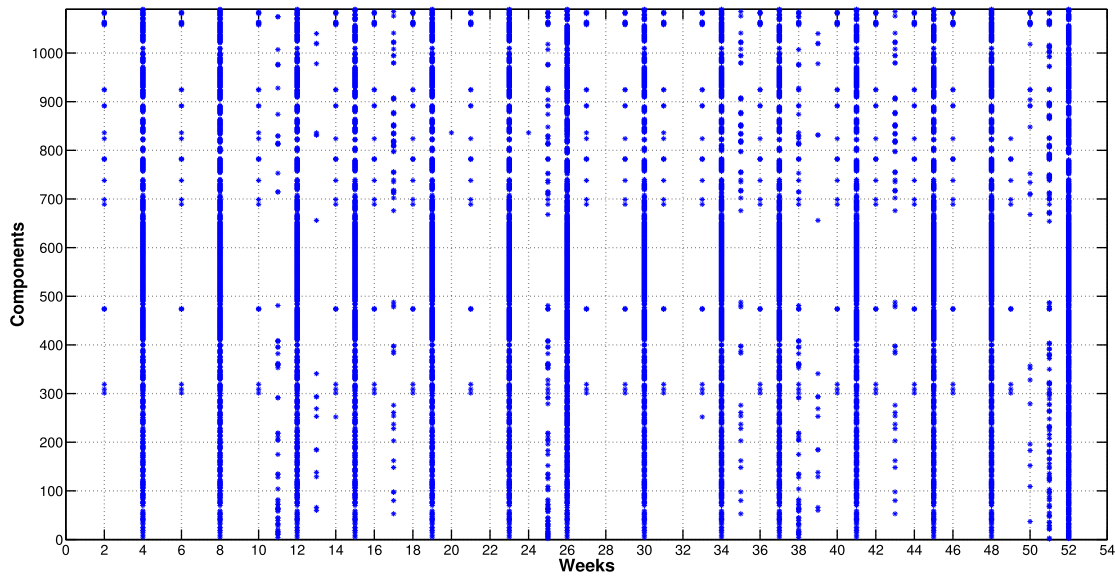


Figure 4. Grouping plan with respect to the maintenance cost objective.

where $C_G^c(GS, PH) = 437,001$ (acu) is the total cost of GM when only the maintenance cost is considered as a unique objective function.

Note that as the workload variance objective is not considered, the grouping plan in Figure 4 shows an imbalance in the number of PM actions assigned each week. For example, the number of PM actions in the 2th week is considerably smaller than that in the 4th. This loss of balance leads to a high variance in the maintenance workload, which equals $V_W^{grouping-c}(GS, K) = 17476$.

Because a high variance in maintenance workload may lead to difficulties in maintenance logistics support and maintenance management, it should be reduced.

7.2.2. Grouping maintenance planning under the workload variance objective

To reduce workload variance, the GA is utilised again to obtain the ideal grouping structure that minimises workload variance. To achieve this, the workload variance weighting factor was set to $\alpha = 1$. First, the convergence of the GA implementation is studied to determine the number of iterations required. The following results were obtained by running the proposed approach for 6000 iterations: $V_W^{grouping-v}(GS, K) = 209$ and $C_G^v(GS, PH) = 474,035$ (acu). When compared to the previous results, in which only the maintenance cost is considered as the unique objective function, it is clear that the workload

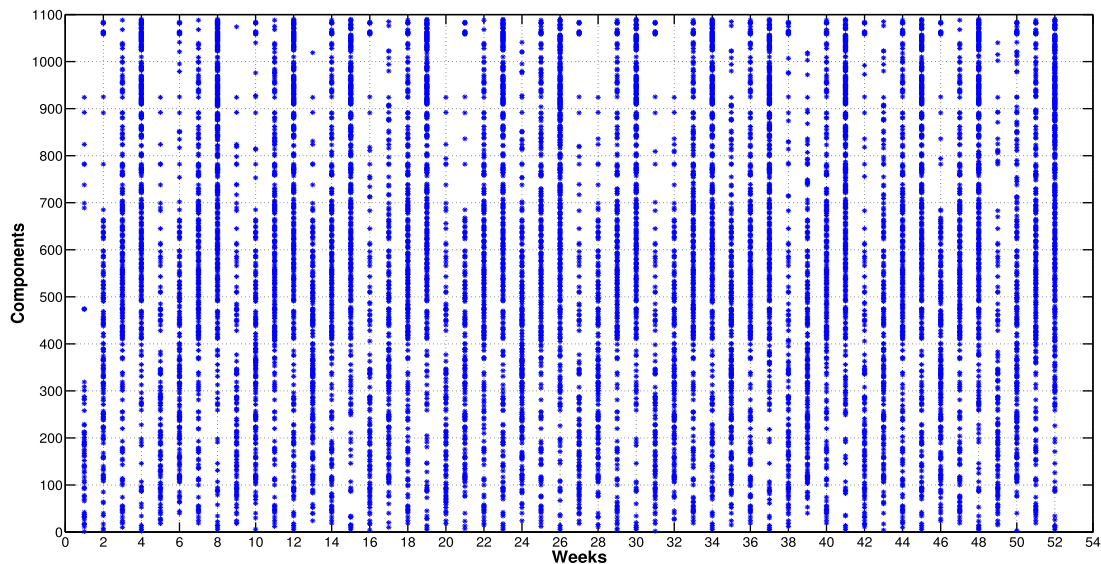


Figure 5. GM plan with respect to cost and variance objectives.

variance is significantly reduced from 17476 to 209 (98.8%). However, maintenance costs increase of

$$\frac{C_G^v(GS, PH) - C_G^c(GS, PH)}{C_G^c(GS, PH)} \cdot 100\% = 8.48\%$$

The maintenance workloads in the planning horizon PH when grouping maintenance is optimised with respect to the workload variance are represented by the blue line in Figure 6(a).

Practically, maintenance logistics support and maintenance management systems may be sufficiently flexible to handle considerable workload variation. Thereafter, we obtain a grouping solution that helps reduce the

Table 2. The workload variance and maintenance cost when α equals 0, 0.5, and 1.

Maintenance models	α	Workload variance	Maintenance cost
IM	–	1,627,000	662,880
Traditional GM	0	17,476	437,001
Bi-objective GM	0.5	337	445,254
Workload GM	1	209	474,035

workload variance but does not lead to a significant increase in the maintenance cost.

7.3. Bi-objective grouping maintenance planning

In this section, the maintenance cost and workload variance are considered objective functions of the optimisation process. Thus, the weighting factor of the workload variance was set between 0 and 1 ($0 < \alpha < 1$). We first study the case where $\alpha = 0.5$, that is, the maintenance cost objective, is as important as the workload variance objective. To determine the number of iterations of the GA, a convergence study was first performed. Appendix 4 provides an illustration of convergence (see Figure A8). Figure 5 illustrates the grouping solution for this case. We can observe that the repartition of PM actions over 52 weeks is more homogeneous than when only the maintenance cost objective is considered (see Figure 4). The maintenance costs and workload variances for $\alpha = 0.5$ are listed in Table 2. The corresponding grouping solution seems reasonable from both the maintenance cost and workload variance viewpoints.

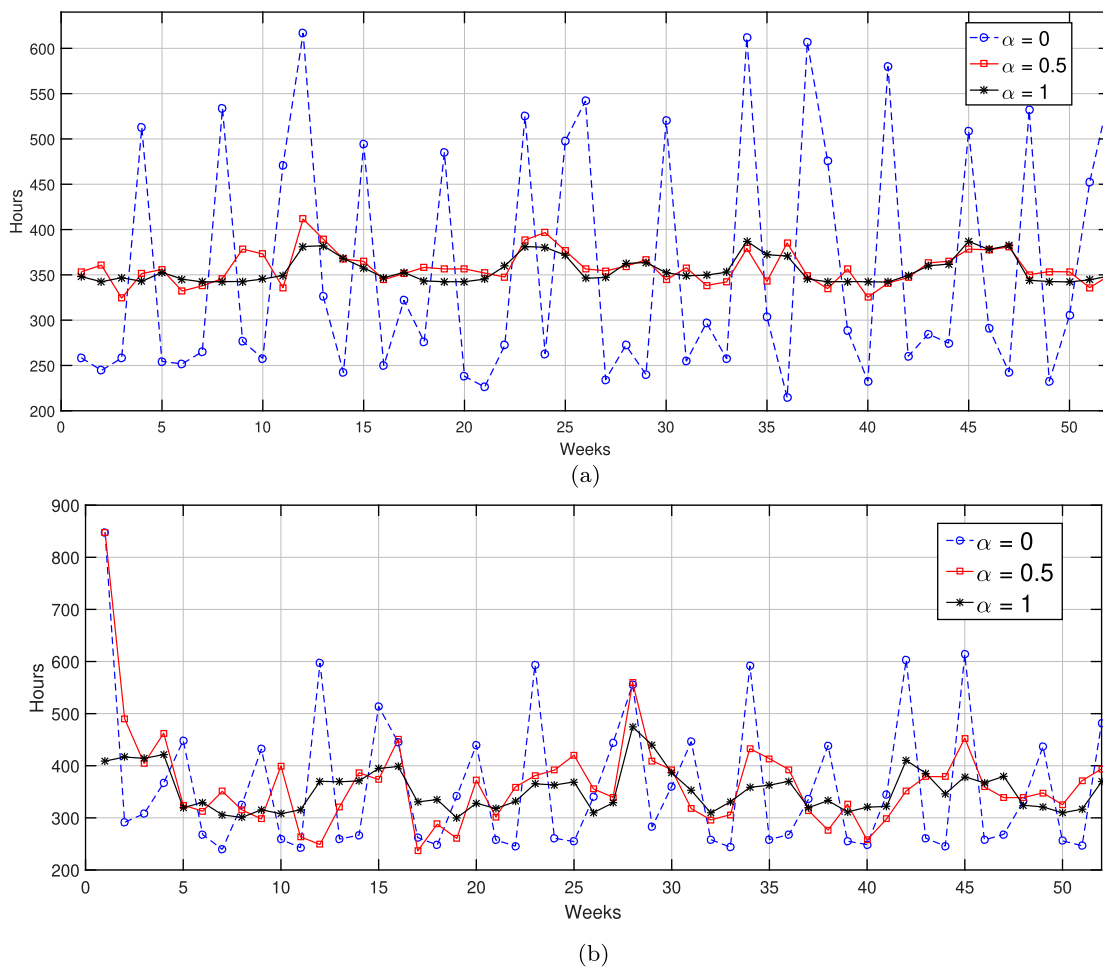


Figure 6. Maintenance workloads when α equals 0, 0.5, and 1 respectively. (a) Constraint (26) is not considered and (b) Constraint (26) is considered.

The maintenance workloads over time in the three cases where α is set to 0, 0.5, and 1 are plotted in Figure 6(a).

Finally, to facilitate decision-making, the Pareto front is constructed by varying the value of α from 0 to 1 in steps $\Delta\alpha = 0.05$. The results are shown in Figure 7(a).

The results show that the higher the maintenance cost, the lower the workload variance. These results allow the maintenance decider to choose an appropriate grouping maintenance plan that satisfies both the workload constraint and maintenance cost.

Considering the adequate performance of our proposed approach in reducing maintenance costs and workload variance, its application to maintenance workshops may face some practical issues, such as the

disordering of the actual maintenance plan. These issues should be addressed to increase the applicability of the proposed approach. It also provides an opportunity to evaluate the performance of our approach under newly restricted conditions. A detailed discussion of this is presented in the following section.

7.4. Grouping maintenance considering the maintenance order

In this subsection, we extend our proposed optimisation model to consider the maintenance order of the PM actions, which is a practical issue. Indeed, if the first PM action of component i of the line j is executed before that of component i' of the line j' in the initial plan (individual

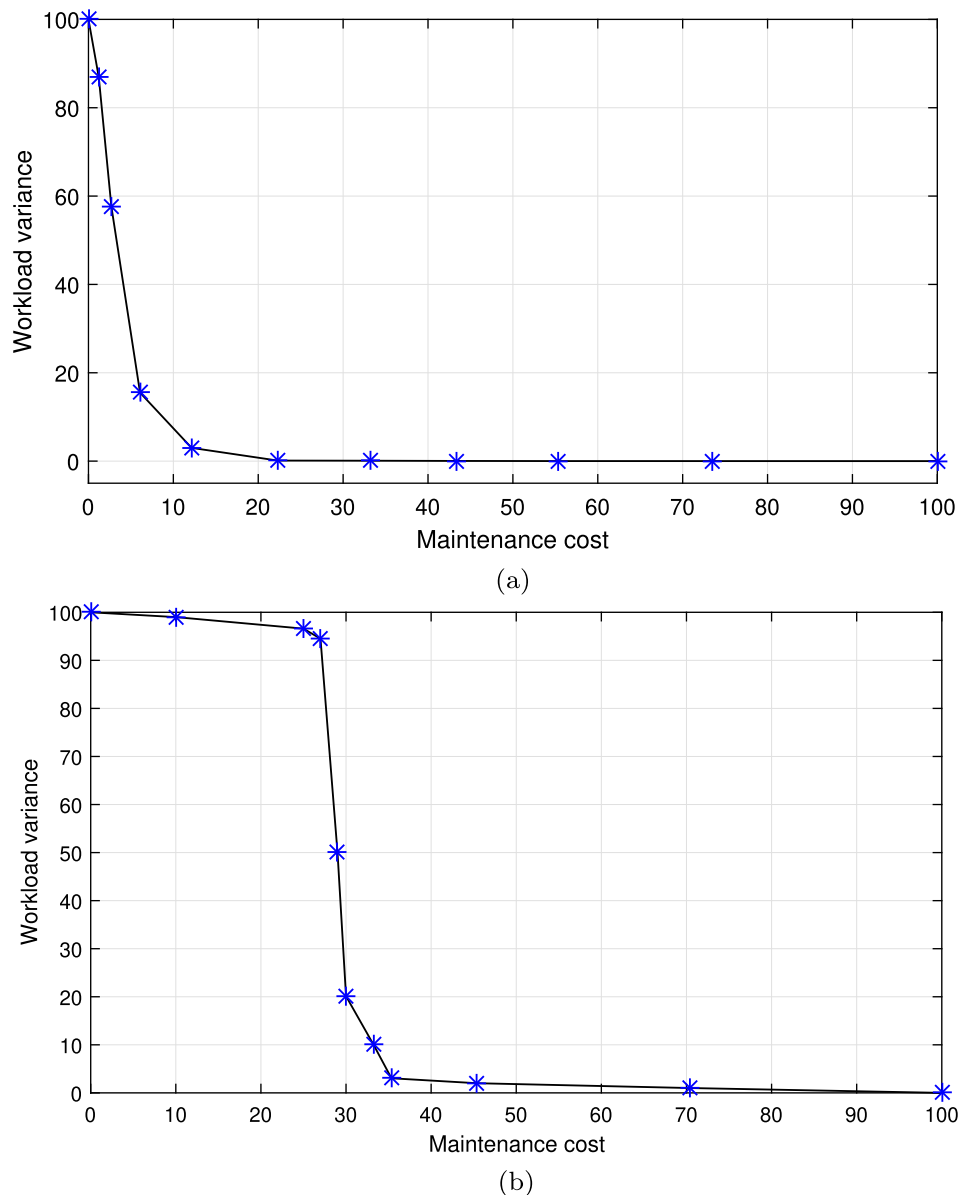


Figure 7. Pareto fronts of the bi-objective grouping optimisation problem. (a) Constraint 26 is not considered and (b) Constraint 26 is considered.

Table 3. Pareto-optimal solutions in the cases where the constraint (26) is added.

α	V_W	C_G	α	V_W	C_G
0	18,482.10 ⁴	59,178.10 ⁵	0.6	16,510.10 ⁴	61,778.10 ⁵
0.1	18,360.10 ⁴	59,299.10 ⁵	0.7	13,849.10 ⁴	63,100.10 ⁵
0.2	18,219.10 ⁴	60,412.10 ⁵	0.8	02,099.10 ⁴	63,198.10 ⁵
0.3	18,009.10 ⁴	61,324.10 ⁵	0.9	01,800.10 ⁴	66,205.10 ⁵
0.4	17,954.10 ⁴	61,485.10 ⁵	1.0	01,726.10 ⁴	90,700.10 ⁵
0.5	16,699.10 ⁴	61,598.10 ⁵	–	–	–

maintenance), i.e. $t_{ij}^1 < t_{i'j'}^1$, the group containing PM PM_{ij}^1 , denoted group G^u , should be maintained before the group containing PM PM_{uv}^1 , denoted group $G^{u'}$, in the GM plan, where G^u and $G^{u'}$ are the groups contain the first PM action of component i of line j and i' of the line j' respectively. Consequently, the disorder problem is solved by introducing the following constraint:

$$t_{G^u} < t_{G^{u'}} \text{ if } \exists \{PM_{ij}^1 \in G^u, PM_{i'j'}^1 \in G^{u'}\} \text{ with } t_{ij}^1 < t_{uv}^1 \quad (26)$$

Figure 6(b) presents the workloads over 52 weeks when constraint (26) was added. The workload variance increased, particularly for $\alpha = 0.5$ and 1. In particular, with $\alpha = 1$, that is, when the workload variance is the most crucial objective, the workload variance is 1726, which is significantly larger than the case in which constraint (26) is not considered ($V_W^{\text{grouping-v}}(\text{GS}, K) = 209$). The results obtained are reasonable because the grouping is less flexible when a constraint is applied.

To obtain the Pareto front, the weighting factor α was varied from 0 to 1 in steps $\Delta\alpha = 0.1$. The results obtained are reported in Table 3 and graphically plotted in Figure 7(b). Note that the maintenance cost and workload variance in this figure are converted into the percentage for better visualisation of the Pareto front. The results show that it is difficult to improve maintenance planning (reduce workload variance and maintenance costs) under the impact of the maintenance order constraint. When α increases from 0 to 1, the workload variance decreases by 90.66% instead of 98.80%, and the maintenance cost increases by 53.27% instead of 8.48%.

8. Conclusions

This study proposes a joint optimisation approach for maintenance grouping and workload balancing. The mathematical formulation of the optimisation problem was derived by considering the economic dependence between components, the specification of both grouping maintenance and workload balancing, and several specific constraints, such as the restricted horizon, maintenance duration, and maintenance orders. A multi-objective optimisation model was then introduced for

maintenance scheduling. Thus, the grouping maintenance plan was optimised with respect to the maintenance cost objective and workload variance. An optimisation algorithm using (Weighted Sum method and a genetic algorithm) was implemented to determine the optimal grouping plan. The performance of the proposed joint optimisation approach was evaluated and validated through a real case study of an automotive plant in France. The results show that our proposed model can significantly reduce both the maintenance cost and workload variance. Additionally, the proposed approach was successfully implemented in several industrial maintenance workshops of an automotive group in France.

In future research, the performance of the GA should be improved given the vast number of PM actions to be considered. Different comparison studies with other multi-objective optimisation meta-heuristics should also be investigated.

Note

1. A partition of a set X , which is defined as a set of nonempty subsets of X such that every element x in X is in exactly one of these subsets (Halmos 2017) (i.e. X is a disjoint union of the subsets)

Data availability statement (DAS)

Due to the nature of this research, participants of this study did not agree for their data to be shared publicly, so supporting data is unavailable.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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References

- Ab-Samat, H., and S. Kamaruddin. 2014. "Opportunistic Maintenance (OM) As a New Advancement in Maintenance Approaches: A Review." *Journal of Quality in Maintenance Engineering* 20 (2): 98–121. <https://doi.org/10.1108/JQME-04-2013-0018>.
- Akl, A. M., S. El Sawah, R. K. Chakraborty, and H. H. Turan. 2022. "A Joint Optimization of Strategic Workforce Planning and Preventive Maintenance Scheduling: A Simulation–Optimization Approach." *Reliability Engineering and System Safety* 219: Article ID 108175. <https://doi.org/10.1016/j.res.2021.108175>.
- Alimian, M., V. Ghezavati, and R. Tavakkoli-Moghaddam. 2020. "New Integration of Preventive Maintenance and Production Planning With Cell Formation and Group Scheduling for Dynamic Cellular Manufacturing Systems." *Journal of Manufacturing Systems* 56:341–358. <https://doi.org/10.1016/j.jmsy.2020.06.011>.
- Bouvard, K., S. Artus, C. Bérenguer, and V. Cocquempot. 2011. "Condition-Based Dynamic Maintenance Operations Planning and Grouping: Application to Commercial Heavy Vehicles." *Reliability Engineering and System Safety* 96 (6): 601–610. <https://doi.org/10.1016/j.res.2010.11.009>.
- Bressi, S., J. Santos, and M. Losa. 2021. "Optimization of Maintenance Strategies for Railway Track-Bed Considering Probabilistic Degradation Models and Different Reliability Levels." *Reliability Engineering and System Safety* 207: Article ID 107359. <https://doi.org/10.1016/j.res.2020.107359>.
- Chalabi, N., M. Dahane, B. Beldjilali, and A. Neki. 2016. "Optimisation of Preventive Maintenance Grouping Strategy for Multi-Component Series Systems: Particle Swarm Based Approach." *Computers and Industrial Engineering* 102:440–451. <https://doi.org/10.1016/j.cie.2016.04.018>.
- Chang, F., G. Zhou, C. Zhang, Z. Xiao, and C. Wang. 2019. "A Service-Oriented Dynamic Multi-level Maintenance Grouping Strategy Based on Prediction Information of Multi-Component Systems." *Journal of Manufacturing Systems* 53:49–61. <https://doi.org/10.1016/j.jmsy.2019.09.005>.
- Chen, G., W. He, L. C. Leung, T. Lan, and Y. Han. 2017. "Assigning Licenced Technicians to Maintenance Tasks At Aircraft Maintenance Base: A Bi-Objective Approach and a Chinese Airline Application." *International Journal of Production Research* 55 (19): 5550–5563. <https://doi.org/10.1080/00207543.2017.1296204>.
- Cohen, A. C. 1965. "Maximium Likelihood Estimation in the Weibull Distribution Based on Complete and on Censored Samples." *Technometrics* 7 (4): 579–588. <https://doi.org/10.1080/00401706.1965.10490300>.
- de Jonge, B., and P. A. Scarf. 2020. "A Review on Maintenance Optimization." *European Journal of Operational Research* 285 (3): 805–824. <https://doi.org/10.1016/j.ejor.2019.09.047>.
- Dekker, R. 1996. "Applications of Maintenance Optimization Models: A Review and Analysis." *Reliability Engineering and System Safety* 51 (3): 229–240. [https://doi.org/10.1016/0951-8320\(95\)00076-3](https://doi.org/10.1016/0951-8320(95)00076-3).
- Dekker, R., H. Frenk, and R. E. Wildeman. 1996. "How to Determine Maintenance Frequencies for Multi-Component Systems? A General Approach." In *Reliability and Maintenance of Complex Systems*, 239–280. New York: Springer-Verlag Berlin Heidelberg.
- Ding, S.-H., and S. Kamaruddin. 2015. "Maintenance Policy Optimization—Literature Review and Directions." *The International Journal of Advanced Manufacturing Technology* 76 (5–8): 1263–1283. <https://doi.org/10.1007/s00170-014-6341-2>.
- Do, P., A. Barros, C. Bérenguer, K. Bouvard, and F. Brissaud. 2013. "Dynamic Grouping Maintenance With Time Limited Opportunities." *Reliability Engineering and System Safety* 120:51–59. <https://doi.org/10.1016/j.res.2013.03.016>.
- Do, P., H. C. Vu, A. Barros, and C. Bérenguer. 2015. "Maintenance Grouping for Multi-Component Systems With Availability Constraints and Limited Maintenance Teams." *Reliability Engineering and System Safety* 142:56–67. <https://doi.org/10.1016/j.res.2015.04.022>.
- Ertogral, K., and F. S. Öztürk. 2019. "An Integrated Production Scheduling and Workforce Capacity Planning Model for the Maintenance and Repair Operations in Airline Industry." *Computers and Industrial Engineering* 127:832–840. <https://doi.org/10.1016/j.cie.2018.11.022>.
- Fan, D., A. Zhang, Q. Feng, B. Cai, Y. Liu, and Y. Ren. 2021. "Group Maintenance Optimization of Subsea Xmas Trees With Stochastic Dependency." *Reliability Engineering and System Safety* 209: Article ID 107450. <https://doi.org/10.1016/j.res.2021.107450>.
- Goyal, S., and M. Kusy. 1985. "Determining Economic Maintenance Frequency for a Family of Machines." *Journal of the Operational Research Society* 36 (12): 1125–1128. <https://doi.org/10.1057/jors.1985.202>.
- Halmos, P. R. 2017. *Naive Set Theory*. Mineola, NY: Courier Dover Publications.
- Hameed, Z., and J. Vatn. 2012. "Role of Grouping in the Development of An Overall Maintenance Optimization Framework for Offshore Wind Turbines." *Proceedings of the Institution of Mechanical Engineers, Part O: Journal of Risk and Reliability* 226 (6): 584–601. <https://doi.org/10.1177/1748006X12464616>.
- Hanini, M., S. Khebbache, L. Bouillaut, and M. Hadji. 2022. "Dynamic and Adaptive Grouping Maintenance Strategies: New Scalable Optimization Algorithms." *Proceedings of the Institution of Mechanical Engineers, Part O: Journal of Risk and Reliability* 236 (5): 647–660. <https://doi.org/10.1177/1748006X211049924>.
- Hashemian, N., C. Diallo, and B. Vizvári. 2014. "Makespan Minimization for Parallel Machines Scheduling With Multiple Availability Constraints." *Annals of Operations Research* 213 (1): 173–186. <https://doi.org/10.1007/s10479-012-1059-8>.

- Jaigirdar, S. M., S. Das, A. R. Chowdhury, S. Ahmed, and R. K. Chakraborty. 2022. "Multi-Objective Multi-Echelon Distribution Planning for Perishable Goods Supply Chain: A Case Study." *International Journal of Systems Science: Operations and Logistics* 10 (1): 1–19. <https://doi.org/10.1080/23302674.2021.2020367>.
- Khatab, A., E. H. Aghezzi, C. Diallo, and I. Djelloul. 2017. "Selective Maintenance Optimisation for Series-Parallel Systems Alternating Missions and Scheduled Breaks With Stochastic Durations." *International Journal of Production Research* 55 (10): 3008–3024. <https://doi.org/10.1080/00207543.2017.1290295>.
- Kim, M. J., and V. Makis. 2009. "Optimal Maintenance Policy for a Multi-State Deteriorating System With Two Types of Failures Under General Repair." *Computers and Industrial Engineering* 57 (1): 298–303. <https://doi.org/10.1016/j.cie.2008.11.023>.
- Kim, I. Y., and O. De Weck. 2006. "Adaptive Weighted Sum Method for Multiobjective Optimization: A New Method for Pareto Front Generation." *Structural and Multidisciplinary Optimization* 31 (2): 105–116. <https://doi.org/10.1007/s00158-005-0557-6>.
- Konak, A., D. W. Coit, and A. E. Smith. 2006. "Multi-Objective Optimization Using Genetic Algorithms: A Tutorial." *Reliability Engineering and System Safety* 91 (9): 992–1007. <https://doi.org/10.1016/j.res.2005.11.018>.
- Koochaki, J., J. A. Bokhorst, H. Wortmann, and W. Klingenberg. 2013. "The Influence of Condition-Based Maintenance on Workforce Planning and Maintenance Scheduling." *International Journal of Production Research* 51 (8): 2339–2351. <https://doi.org/10.1080/00207543.2012.737944>.
- Li, H., E. Deloux, and L. Dieulle. 2016. "A Condition-Based Maintenance Policy for Multi-Component Systems With Lévy Copulas Dependence." *Reliability Engineering and System Safety* 149:44–55. <https://doi.org/10.1016/j.res.2015.12.011>.
- Li, F., Y. Sun, L. Ma, and J. Mathew. 2011. "A Grouping Model for Distributed Pipeline Assets Maintenance Decision." In *Quality, Reliability, Risk, Maintenance, and Safety Engineering (ICQR2MSE), 2011 International Conference on*, 601–606. IEEE.
- Liang, Z., and A. K. Parlikad. 2020. "Predictive Group Maintenance for Multi-System Multi-Component Networks." *Reliability Engineering and System Safety* 195:Article ID 106704. <https://doi.org/10.1016/j.res.2019.106704>.
- Lu, Y., L. Sun, J. Kang, H. Sun, and X. Zhang. 2017. "Opportunistic Maintenance Optimization for Offshore Wind Turbine Electrical and Electronic System Based on Rolling Horizon Approach." *Journal of Renewable and Sustainable Energy* 9 (3): Article ID 033307. <https://doi.org/10.1063/1.4989640>.
- Luiz Junho Pereira, J., G. Antônio Oliver, M. Brendon Francisco, S. Simões Cunha Jr, and G. Ferreira Gomes. 2022. "Multi-objective Lichtenberg Algorithm: A Hybrid Physics-based Meta-heuristic for Solving Engineering Problems." *Expert Systems with Applications* 187:Article ID 115939. <https://doi.org/10.1016/j.eswa.2021.115939>.
- Marler, R. T., and J. S. Arora. 2010. "The Weighted Sum Method for Multi-Objective Optimization: New Insights." *Structural and Multidisciplinary Optimization* 41 (6): 853–862. <https://doi.org/10.1007/s00158-009-0460-7>.
- Nakagawa, T., and M. Kowada. 1983. "Analysis of a System With Minimal Repair and Its Application to Replacement Policy." *European Journal of Operational Research* 12 (2): 176–182. [https://doi.org/10.1016/0377-2217\(83\)90221-7](https://doi.org/10.1016/0377-2217(83)90221-7).
- Nakagawa, T., and S. Mizutani. 2009. "A Summary of Maintenance Policies for a Finite Interval." *Reliability Engineering & System Safety* 94 (1): 89–96. <https://doi.org/10.1016/j.res.2007.04.004>.
- Nguyen, H. S. H., P. Do, H.-C. Vu, and B. Iung. 2019. "Dynamic Maintenance Grouping and Routing for Geographically Dispersed Production Systems." *Reliability Engineering and System Safety* 185:392–404. <https://doi.org/10.1016/j.res.2018.12.031>.
- Nicolai, R. P., and R. Dekker. 2008. "Optimal Maintenance of Multi-Component Systems: A Review." In *Complex System Maintenance Handbook*, 263–286. London: Springer-Verlag.
- Noortwijk, J. v., R. Dekker, R. M. Cooke, and T. Mazucchi. 1992. "Expert Judgement in Maintenance Optimization." *IEEE Transactions on Reliability*, 427–432. <https://doi.org/10.1109/24.159813>.
- Nowakowski, T., and S. Werbińska. 2009. "On Problems of Multicomponent System Maintenance Modelling." *International Journal of Automation and Computing* 6 (4): 364–378. <https://doi.org/10.1007/s11633-009-0364-4>.
- Okoh, P. 2015. "Maintenance Grouping Optimization for the Management of Risk in Offshore Riser System." *Process Safety and Environmental Protection* 98:33–39. <https://doi.org/10.1016/j.psep.2015.06.007>.
- Pargar, F., O. Kauppila, and J. Kujala. 2017. "Integrated Scheduling of Preventive Maintenance and Renewal Projects for Multi-Unit Systems With Grouping and Balancing." *Computers and Industrial Engineering* 110:43–58. <https://doi.org/10.1016/j.cie.2017.05.024>.
- Rebaiaia, M. L., and D. Ait-kadi. 2021. "Maintenance Policies With Minimal Repair and Replacement on Failures: Analysis and Comparison." *International Journal of Production Research* 59 (23): 6995–7017. <https://doi.org/10.1080/00207543.2020.1832275>.
- Safaei, N., D. Banjevic, and A. K. Jardine. 2011. "Workforce-Constrained Maintenance Scheduling for Military Aircraft Fleet: A Case Study." *Annals of Operations Research* 186 (1): 295–316. <https://doi.org/10.1007/s10479-011-0885-4>.
- Sharma, R., H. Kodamana, and M. Ramteke. 2022. "Multi-Objective Dynamic Optimization of Hybrid Renewable Energy Systems." *Chemical Engineering and Processing-Process Intensification* 170:Article ID 108663. <https://doi.org/10.1016/j.cep.2021.108663>.
- Sharma, A., G. Yadava, and S. Deshmukh. 2011. "A Literature Review and Future Perspectives on Maintenance Optimization." *Journal of Quality in Maintenance Engineering* 17 (1): 5–25. <https://doi.org/10.1108/13552511111116222>.
- Sheikhalishahi, M., A. Azadeh, and L. Pintelon. 2017. "Dynamic Maintenance Planning Approach by Considering Grouping Strategy and Human Factors." *Process Safety and Environmental Protection* 107:289–298. <https://doi.org/10.1016/j.psep.2017.02.015>.
- Tavakoli Kafiabad, S., M. K. Zanjani, and M. Nourelfath. 2022. "Workforce Training and Operations Planning for Maintenance Centres Under Demand Uncertainty." *International Journal of Production Research* 60 (5): 1587–1599. <https://doi.org/10.1080/00207543.2020.1866781>.

- Thoman, D. R., L. J. Bain, and C. E. Antle. 1969. "Inferences on the Parameters of the Weibull Distribution." *Technometrics* 11 (3): 445–460. <https://doi.org/10.1080/00401706.1969.10490706>.
- Thomas, E., E. Levrat, and B. Iung. 2008. "Overview on Opportunistic Maintenance." *IFAC Proceedings Volumes* 41 (3): 245–250. <https://doi.org/10.3182/20081205-2-CL-4009.00044>.
- Turan, H. H., F. Kusanoglu, and M. Atmis. 2021. "A Multi-Skilled Workforce Optimisation in Maintenance Logistics Networks by Multi-Thread Simulated Annealing Algorithms." *International Journal of Production Research* 59 (9): 2624–2646. <https://doi.org/10.1080/00207543.2020.1735665>.
- van Dijkhuizen, G. 2000. "Maintenance Grouping in Multi-Step Multi-Component Production Systems." In *Maintenance, Modeling and Optimization*, 283–306. New York: Springer Science+Business Media.
- Van Horenbeek, A., and L. Pintelon. 2013. "A Dynamic Predictive Maintenance Policy for Complex Multi-Component Systems." *Reliability Engineering and System Safety* 120:39–50. <https://doi.org/10.1016/j.res.2013.02.029>.
- Van Tung, T., and B.-S. Yang. 2009. "Machine Fault Diagnosis and Prognosis: The State of the Art." *International Journal of Fluid Machinery and Systems* 2 (1): 61–71. <https://doi.org/10.5293/IJFMS.2009.2.1.061>.
- Vatn, J., P. Hokstad, and L. Bodsberg. 1996. "An Overall Model for Maintenance Optimization." *Reliability Engineering and System Safety* 51 (3): 241–257. [https://doi.org/10.1016/0951-8320\(95\)00055-0](https://doi.org/10.1016/0951-8320(95)00055-0).
- Viveros, P., R. Mena, E. Zio, L. Miqueles, and F. Kristjanpoller. 2021. "Integrated Planning Framework for Preventive Maintenance Grouping: A Case Study for a Conveyor System in the Chilean Mining Industry." *Proceedings of the Institution of Mechanical Engineers, Part O: Journal of Risk and Reliability*, Article ID 1748006X211053758. <https://doi.org/10.1177/1748006X211053758>.
- Vu, H. C., P. Do, and A. Barros. 2015. "A Stationary Grouping Maintenance Strategy Using Mean Residual Life and the Birnbaum Importance Measure for Complex Structures." *IEEE Transactions on Reliability* 65 (1): 217–234. <https://doi.org/10.1109/TR.2015.2455498>.
- Vu, H. C., P. Do, and A. Barros. 2018. "A Study on the Impacts of Maintenance Duration on Dynamic Grouping Modeling and Optimization of Multicomponent Systems." *IEEE Transactions on Reliability* 67 (3): 1377–1392. <https://doi.org/10.1109/TR.2018.2827926>.
- Vu, H. C., P. Do, A. Barros, and C. Bérenguer. 2014a. "Maintenance Grouping Strategy for Multi-Component Systems With Dynamic Contexts." *Reliability Engineering and System Safety* 132:233–249. <https://doi.org/10.1016/j.res.2014.08.002>.
- Vu, H. C., P. Do, A. Barros, and C. Bérenguer. 2014b. "Maintenance Planning and Dynamic Grouping for Multi-Component Systems With Positive and Negative Economic Dependencies." *IMA Journal of Management Mathematics* 26 (2): 145–170. <https://doi.org/10.1093/imaman/dpu007>.
- Vu, H. C., P. Do, M. Fouladirad, and A. Grall. 2020. "Dynamic Opportunistic Maintenance Planning for Multi-Component Redundant Systems With Various Types of Opportunities." *Reliability Engineering and System Safety* 198:Article ID 106854. <https://doi.org/10.1016/j.res.2020.106854>.
- Wang, S., and M. Liu. 2014. "Two-Stage Hybrid Flow Shop Scheduling with Preventive Maintenance Using Multi-Objective Tabu Search Method." *International Journal of Production Research* 52 (5): 1495–1508. <https://doi.org/10.1080/00207543.2013.847983>.
- Wildeman, R. E., R. Dekker, and A. Smit. 1997. "A Dynamic Policy for Grouping Maintenance Activities." *European Journal of Operational Research* 99 (3): 530–551. [https://doi.org/10.1016/S0377-2217\(97\)00319-6](https://doi.org/10.1016/S0377-2217(97)00319-6).
- Wu, J., T. S. A. Ng, M. Xie, and H.-Z. Huang. 2010. "Analysis of Maintenance Policies for Finite Life-cycle Multistate Systems." *Computers and Industrial Engineering* 59 (4): 638–646. <https://doi.org/10.1016/j.cie.2010.07.013>.
- Zhou, A., B.-Y. Qu, H. Li, S.-Z. Zhao, P. N. Suganthan, and Q. Zhang. 2011. "Multiobjective Evolutionary Algorithms: A Survey of the State of the Art." *Swarm and Evolutionary Computation* 1 (1): 32–49. <https://doi.org/10.1016/j.swevo.2011.03.001>.

Appendices

Appendix 1. Assignments of the PM tasks using the MILP

Here, we explain how to assign n_{G^u} PM tasks to N_r repairmen. To prioritise the parallel execution of PM tasks of the same production line, the assignment is performed repeatedly for only the PM tasks of each production line ($t_{ij}^l \in G_j^u$), instead of for all PM tasks of the group ($t_{ij}^l \in G^u$). Note that repairs may not be available at the same time, depending on the assigned PM tasks of the previous production line. The MILP model proposed in Hashemian, Diallo, and Vizvári (2014) for makespan minimisation with unavailable constraints is adapted.

For more details, let r ($r = 1, \dots, N_r$) and p ($p = 1, \dots, n_{uj}$) denote the repairmen and PM task indices, respectively. n_{uj} is the number of PM tasks in line j in group G^u .

$$\begin{aligned}
 \min_{x,y} \quad & C_{\max} \\
 \text{s.t.} \quad & \sum_{p=1}^{n_{uj}} d_p x_{rp} \leq s_r y_r + M(1 - y_r), \quad r = 1, \dots, N_r \\
 & \sum_{p=1}^{n_{uj}} d_p x_{rp} + (e_r - s_r)(1 - y_r) \leq C_{\max}, \quad r = 1, \dots, N_r \\
 & \sum_{r=1}^{N_r} x_{rp} = 1, \quad p = 1, \dots, n_{uj}
 \end{aligned} \tag{A1}$$

where $C_{\max}$ is the makespan, s_r and e_r denote the start and end of the non-availability period of repairman r , respectively, and M is an appropriately large number.

$$x_{rp} = \begin{cases} 1 & \text{if PM task } p \text{ is assigned to repairman } r \\ 0 & \text{otherwise} \end{cases} \tag{A2}$$

$$y_r = \begin{cases} 1 & \text{if PM tasks assigned to repairman } r \\ & \text{are completed before } s_r \\ 0 & \text{otherwise} \end{cases} \tag{A3}$$

The problem was solved directly using the optimisation toolbox in MATLAB. Finally, the maintenance order of the production lines can be chosen in accordance with the production planning, availability of spare parts, etc.

Appendix 2. Normalization of the workload variance and maintenance cost values

The normalised value of the workload variance $\tilde{V}_W(GS, K)$ is calculated as:

$$\tilde{V}_W(GS, K) = \frac{V_W^{\max} - V_W}{V_W^{\max} - V_W^{\min}} \quad (A4)$$

where $V_W^{\max}$ and $V_W^{\min}$ are the maximum and minimum values of workload variance, respectively. These two values can be obtained by considering the same problems as those in Equations (21a)–(21f) except that the second objective (maintenance cost) is removed.

$$GS_V^{\max} = \arg \max_{GS} V_W(GS, K) \quad \text{subject to } 20b - 20g \quad (A5)$$

$$V_W^{\max} = V_W(GS_V^{\max}, K)$$

and

$$GS_V^{\min} = \arg \min_{GS} V_W(GS, K) \quad \text{subject to } 20b - 20g \quad (A6)$$

$$V_W^{\min} = V_W(GS_V^{\min}, K)$$

A genetic algorithm is used to solve the mono-objective optimisation problem in Equations (A5) and (A6). Similarly, the maintenance cost can be normalised as:

$$\tilde{C}_G(GS, PH) = \frac{C_G^{\max} - C_G}{C_G^{\max} - C_G^{\min}} \quad (A7)$$

where $C_G^{\max}$ and $C_G^{\min}$ are the maximum and minimum maintenance costs, respectively. The determination of $C_G^{\min}$ is similar to that of $V_W^{\max}$ or $V_W^{\min}$, and is performed by solving the following mono-objective optimisation problem:

$$GS_C^{\min} = \arg \min_{GS} C_G(GS, PH) \quad \text{subject to } 20b - 20g$$

$$C_G^{\min} = C_G(GS_C^{\min}, PH) \quad (A8)$$

In contrast, the maintenance cost is maximal when maintenance activities are conducted separately. $C_G^{\max}$ can be directly calculated using Equation (14).

$$C_G^{\max} = C_I(PH) = d_{PH} \cdot \sum_{j=1}^N \sum_{i=1}^{n_j} CR_{ij}(x_{ij}^*) \quad (A9)$$

Appendix 3. Implementation of GA

The GA is a powerful and flexible tool for searching for the global optimum. It can be applied to diverse optimisation problems with and without constraints. The GA begins by randomly creating a set (population) of solutions. These solutions are then evaluated by their fitness function and modified by GA operators (crossover and mutation). The evaluation and modification processes are stopped when at least one of the stopping conditions is satisfied. Detailed descriptions of the main steps of the GA when applied to our optimisation problem are presented in the following paragraphs.

A.1 Coding

This step determines the method of introducing a solution in GA. For our grouping optimisation problem, a grouping structure is coded by an array of M elements denoted X . Array X is constructed using the following encoding function:

$$PM_{ij}^l \in G^u \Rightarrow X(PM_{ij}^l) = y; \quad y \in [1, M] \quad (A10)$$

A.2 Generation of initial population

As mentioned above, GA begins by creating a set of solutions (the initial population). If there are few solutions in the population, only a small portion of the research space is explored. However, if there are significantly many grouping structures in the initial population, the GA slows. Considering that many PM actions should be assumed in our case study, the population size was set to 200 solutions. The number of groups in the solution was randomly chosen between 1 and M to generate a solution. Next, all PM actions were randomly placed into chosen groups such that each group contained at least one PM action.

A.3 Calculation of fitness function

Each solution is evaluated by its fitness function defined in Equation (24).

$$\tilde{VC}(GS, K, PH) = \alpha \cdot \tilde{V}_W(GS, K) + (1 - \alpha) \cdot \tilde{C}_G(GS, PH)$$

The lower the value of the fitness function, the better the solution. In this step, the constraints (21b)–(21f) or (26) are considered. Specifically, if the grouping structure GS does not satisfy one of these constraints, its fitness value $\tilde{VC}(GS)$ is set to a random value larger than 1. Considering that the maximum value of $\tilde{VC}$ is 1, an unfeasible solution (with a fitness value greater than 1) is automatically eliminated from the population after several iterations.

A.4 Selection to crossover

The objective is to select a pair of solutions that will be considered in the crossover step. The selection process begins by sorting all the population solutions according to their fitness function values in ascending order. The population is divided into s groups (the first group contains M/s best solutions). Subsequently, a solution was randomly selected from among these groups. Let p_u denote the probability that a solution of group u is selected, and let $p_1 > p_2 > \dots > p_s$ and $\sum_{i=1}^s p_i = 1$. The objective is to ensure that these good solutions are likely to be selected for the crossover step. The population was divided into 5 groups of 40 solutions for the problem. The group selection probabilities are $p_1 = 0.50, p_2 = 0.2, p_3 = 0.15, p_4 = 0.10, p_5 = 0.05$.

A.5 Crossover

The goal is to combine the selected solutions to generate new child solutions by preserving specific characteristics of their parents. The crossover probability is set to 80%. The one-point crossover was used in our study. A break-point was randomly selected to separate each parent solution into two parts. Subsequently, two-child solutions were generated by exchanging the two parts between the parent solutions.

A.6 Mutation

The mutation is conducted to prevent the GA from falling into local optimal solutions, but it cannot be done often because the GA will change to a random search. Therefore, the probability of a mutation is chosen to be between 0.1% and 5%. The mutation probability was set to 5% for the first 1500 iterations, and 0.1% for the last 500 iterations. The GA was stopped after 2000 iterations. To ensure the convergence of the GA, the ideal grouping solution was only recorded after 5 GA runs.

Appendix 4. Illustration of the convergence

Figure A8 shows the convergence of the overall objective when $\alpha = 0.5$. According to these figures, the proposed algorithm surely converges after 4000 iterations.

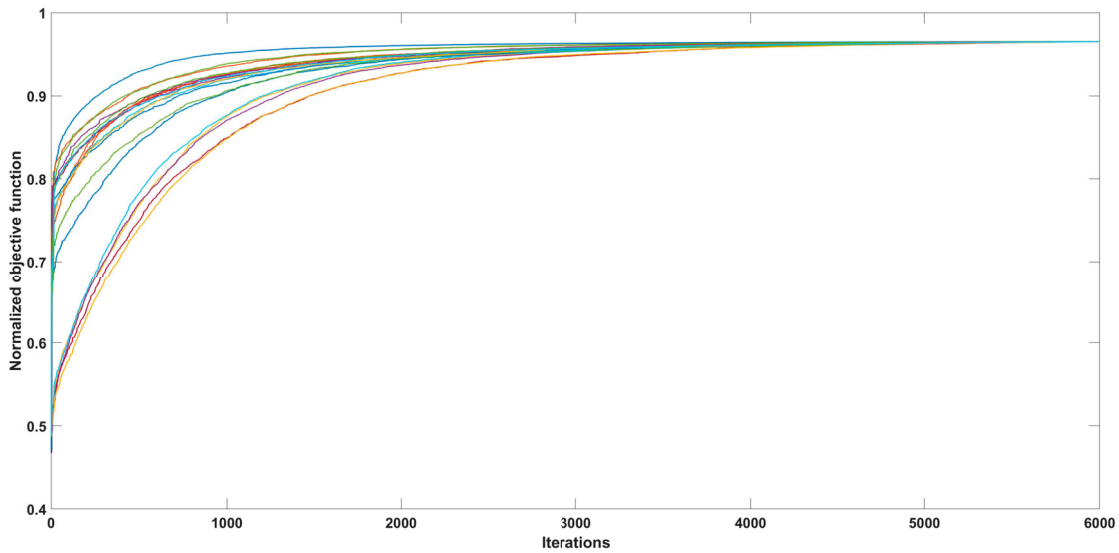


Figure A8. Illustration of the convergence of WSGA with $\alpha = 0.5$.

Appendix 5. A comparison between the WSGA and an exact method

To evaluate the performance of the WSGA, a small-scale problem was considered to obtain the exact solution. The number of machines used for the small-scale problem was 15, as recommended by Vu et al. (2014a). The results validated the performance of the WSGA because the maintenance plan provided by the WSGA was the same as that provided by the exact algorithm. The maintenance cost and workload variance of the 15-machines problem were 173 and 34,324, respectively, for $\alpha = 0.5$.